

ADITYA VISHWAKARMA

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EDUCATION

Indian Institute of Technology Guwahati

Bachelor of Technology in Engineering Physics

Aug 2024 – Present

CGPA: 7.12

PROJECTS

Open Food Facts (Dart SDK) | Open Source | PR #1215 | PR #1212

Mar 2026 – May 2026

- Created a line-by-line **TSV parser** with **O(L)** space complexity, processing **500MB (4.37M rows)** in **9.8s**.
- Validated column headers and added unit tests, achieving a **99.96%** parsing success rate across all datasets.
- Fixed legacy crashes by migrating **product_quantity** from **int** to **num** to safely handle decimal values.
- Delivered backward-compatible **@deprecated** bridges for **quantityString** to ensure zero downtime for clients.

Vehicle Route Optimization App | Kriti 2026 | bit.ly/velora_opti

Jan 2026 – Mar 2026

- Designed a cross-platform **Flutter** application with **Supabase** backend for complex optimization workflows.
- Rendered route visualizations and analytics dashboards using **Leaflet**, **fl_chart**, and **ValueNotifier**.
- Automated multi-format data export pipelines to generate **JSON**, **Excel**, and **PDF** reports seamlessly.
- Reduced route costs on MT-CVRPTW benchmarks using a custom **C++17 ALNS** solver with **Flask** APIs and **OpenRouteService**.

Out of Sight — 3D Stealth Game | Kriti 2026 | bit.ly/GDAI-Gamejam

Feb 2026 – Mar 2026

- Programmed a 3D third-person stealth game on **Godot 4.5** integrating **FSM-driven** enemy AI architectures.
- Implemented **NavMesh** pathfinding, **RayCast3D** vision detection, and stamina movement via **Jolt Physics**.
- Wrote custom **GLSL** shaders for player invisibility effects and procedural **skybox** rendering systems.
- Optimized runtime performance using pre-baked **Terrain3D** NavMeshes and throttled path recalculation.

IITGCampusSync | Coding Club IITG | bit.ly/IITGCampusSync

Aug 2025 – Oct 2025

- Developed a personalized tag recommendation engine using **MongoDB** and **Mongoose** schemas.
- Modeled many-to-many data relationships across collections supporting concurrent campus users.
- Exposed production-grade **REST API** endpoints with **Node.js** and **Express.js** for event management.
- Enforced referential integrity across collections using atomic **MongoDB \$pull** and **\$addToSet** operations.

SKILLS

Languages: C++, Dart, JavaScript, Python, C#, GDScript, C, HTML, CSS

Technologies: Flutter, Node.js, Express.js, Supabase, MongoDB, PostgreSQL, Godot, Unity

Tools & Workflows: Git, GitHub, Postman, Docker

POSITION OF RESPONSIBILITY

Student's Web Committee (SWC), IIT Guwahati | Senior App Developer

Apr 2026 – Present

- Lead application development, optimized code architectures, and mentored developers for campus apps.

GameDev & Esports Club, IIT Guwahati | Game Developer

Jun 2025 – Apr 2026

- Shipped game projects in Godot and Unity, contributing to club showcases and inter-college competitions.

Umiam Hostel, IIT Guwahati | Associate Welfare Secretary

Apr 2025 – Apr 2026

- Coordinated welfare events for hostel residents, managing budgets and inter-hostel activity scheduling.

KEY ACHIEVEMENTS

- Route Optimization Challenge:** Secured **4th Place** at Kriti, IIT Guwahati. 2026
- Codeforces:** Maximum Rating - **1402** | Rated **Specialist** | Handle — *tinyurl.com/adityav27*
- CodeChef:** Maximum Rating - **1511** | Rated **2 Stars** | Handle — *genie_craft_28*
 - Codechef Starters 241 (Div 4) : Secured **Global Rank 575** among **15,000+** participants. 2026
- WorldQuant IQC 2025 – Stage 1:** Rank **2,259** out of **77,000+** teams worldwide. 2025
- JEE Mains & Advanced:** Secured **99.35 percentile** in JEE Mains and ranked among the **top 6.55%** out of 1.86 lakh candidates in JEE Advanced. 2024
- IISER Aptitude Test (IAT):** Secured **AIR 1296** among ~35,000 candidates nationwide (Top ~4%). 2024